

Hypothesis Tests and Control Charts

Module 5

Subject Name: Statistics and
Probability

Subject Code: BAS0303N



Dr. Lokesh Chaudhary
Department of Mathematics

B Tech 3rd Sem

Module 1: (Statistical Techniques-I)

Introduction: Measures of central tendency: Mean, Median, Mode, Standard deviation, Quartile deviation, Moment, Skewness, Kurtosis.

Module 2: (Statistical Techniques-II)

Curve Fitting, Method of least squares, fitting of straight lines, Fitting of second-degree parabola, Exponential curves, Correlation and Rank correlation, Linear regression, nonlinear regression and multiple linear regression.

Module 3: (Probability and Random Variable)

Random Variable: Definition of a Random Variable, Discrete Random Variable, Continuous Random Variable, Probability mass function, Probability Density Function, Distribution functions.

Multiple Random Variables: Joint density and distribution Function, Properties of Joint Distribution function, Marginal density Functions, Conditional Distribution and Density, Statistical Independence, Central Limit Theorem (Proof not expected).

Module 4: (Expectations and Probability Distribution)

Expectations of single Random Variable, Mean, Variance, Moment Generating Function, Binomial, Poisson, Normal, Exponential distribution.

Module 5: (Hypothesis Tests and Control Charts)

Testing a Hypothesis, Null hypothesis, Alternative hypothesis, Level of significance, Confidence limits, Test of significance of difference of means, Z-test, t-test and Chi-square test, F-test, One way ANOVA.

Statistical Quality Control (SQC), Control Charts, Control Charts for variables (Mean and Range Charts), Control Charts for Variables (p, np and C charts).

Branch Wise Application

- ❖ Data Analysis
- ❖ Artificial intelligence
- ❖ Network and Traffic modeling

Course Objectives

- The objective of this course is to familiarize the students with concepts of Probability and statistical techniques. It aims to equip the students with adequate Knowledge of statistics that will enable them in formulating Problems and solving problems analytically.

The students will learn:

- Understand the concept of correlation, moments, skewness and kurtosis and curve fitting.
- Apply the concept of hypothesis testing and statistical quality control to create control charts.
- Remember the concept of probability to evaluate probability distributions.
- Understand the concept of Mathematical Expectations and Probability Distribution.

CO1: Apply the concept of moments, skewness and kurtosis in relevant field.

CO2: Apply the concept of correlation, regression and curve fitting with real world problems.

CO3: Apply the concept of probability and random variable.

CO4: Apply the concept of Mathematical Expectations and Probability Distribution in real life problems.

CO5: Apply the concept of hypothesis testing and statistical quality control to create control charts.

Prerequisite and Recap

- Knowledge of Maths 1 B.Tech.
- Knowledge of Maths 2 B.Tech.
- Knowledge of Permutation and Combination.

- Testing a Hypothesis,
- Null and Alternative hypothesis
- Level of significance
- Confidence limits
- p-value
- Test of significance of difference of means:
- Z-test
- t-test
- Chi-square test
- F-test
- ANOVA: One way and Two way
- Statistical Quality Control (SQC)
- Control Charts
- Control Charts for variables (Mean and Range Charts)
- Control Charts for Variables ($\bar{p}$, np and C charts).

Module Objective(CO5)

- The objective of this course is to familiarize the engineers with Statistical techniques and its applications in real world.
- The concept of hypothesis, ANOVA and quality control by control charts.

Test of Hypothesis

- Describe real-life examples to explain the motivation behind hypothesis

Test of Hypothesis:

- Some information about a characteristic of the population is known, we wish to know whether the information can be accepted or rejected, we choose a random sample and obtain information about the characteristic, based on this information.
- we conclude whether the available information of the characteristic of the population can be accepted or rejected. This is called Test of hypothesis.

Symbols for Population & Samples:

Symbols	Population parameter/ characteristic	Sample Statistic
	Population Size (N)	Sample Size (n)
	Population Mean (μ)	Sample Mean ($\bar{x}$)
	Population Standard Deviation (σ)	Sample Standard Deviation (s)
	Population Variance (σ^2)	Sample Variance (s^2)

Procedure of Test of hypothesis

- State Null hypothesis & Alternate hypothesis
- Define level of significance
- Use suitable test statistics
- Establish critical region (Table value)
- Conclusion

Null Hypothesis

- A population is given to us and we wish to have information about a parameter of the Population.
- we start with the assumption that there is no significant difference between the sample statistic and the corresponding population parameter or between two sample statistic.
- This assumption there is no significant difference is called a null hypothesis and is denoted by H_0 .

Alternate Hypothesis

A hypothesis that is different from the null hypothesis is called alternate hypothesis and is denoted by H_1 .

Single tailed & two tailed test

Consider the following example-

let the null hypothesis is define as-

H_0 : The population has an assume value of mean μ_0 i.e. $\mu = \mu_0$.

Then alternate hypothesis can be define as any of the following-

I. $H_1: \mu \neq \mu_0$

II. $H_1: \mu > \mu_0$

III. $H_1: \mu < \mu_0$

Depending upon the problem we use single tail or two tailed test to estimate the significance of result. In two tailed test the areas of the both the tails of the curve representing the sampling distribution are taken into account whereas in single tail test only the area on left or right of is taken into account.

Level of significance

The probability level below which we reject the hypothesis is known as level of significance. It is denoted by α and generally specified before sample are drawn.

Generally we take $\alpha = 5\%$ or 1% .

Critical value

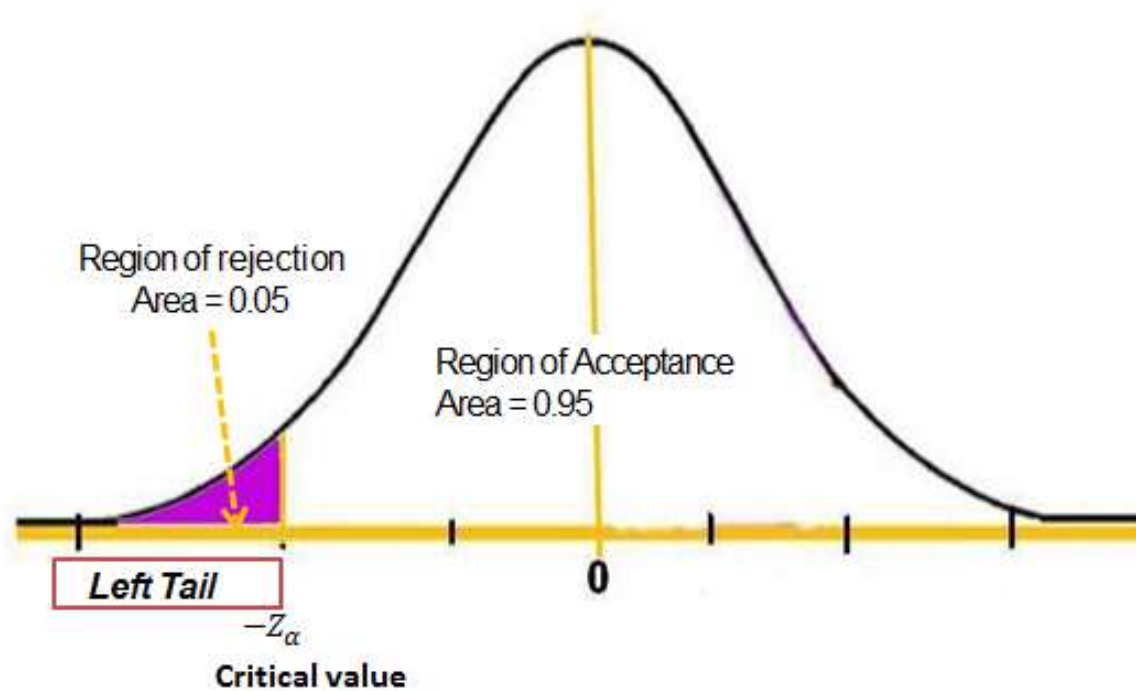
The value of test statistic which separates the critical region (or rejection) and acceptance region is called the critical value and it depends upon

- Level of significance
- Alternate hypothesis (whether it is two tailed or single tailed)

Critical Region or rejection region

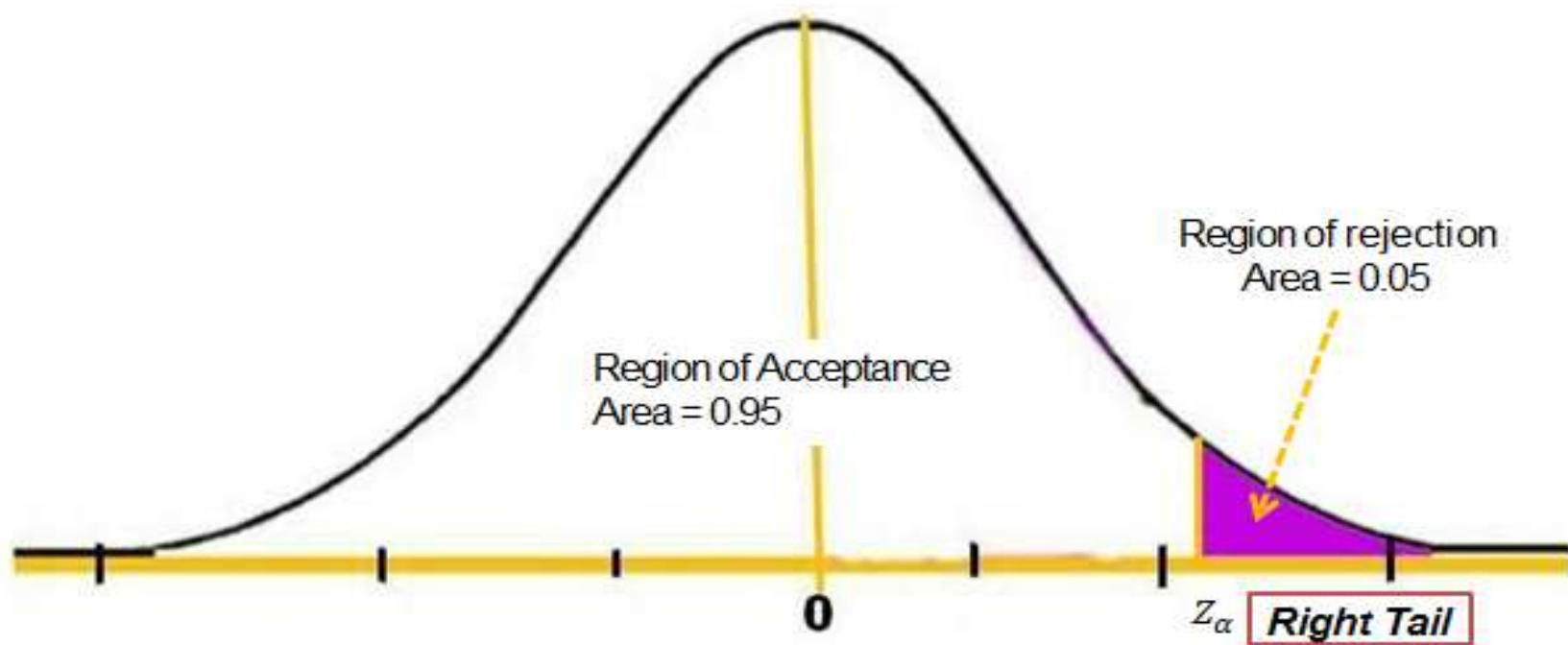
Critical region is the region in which a sample value is rejected.

Single left tailed test $\alpha = 5\%$:

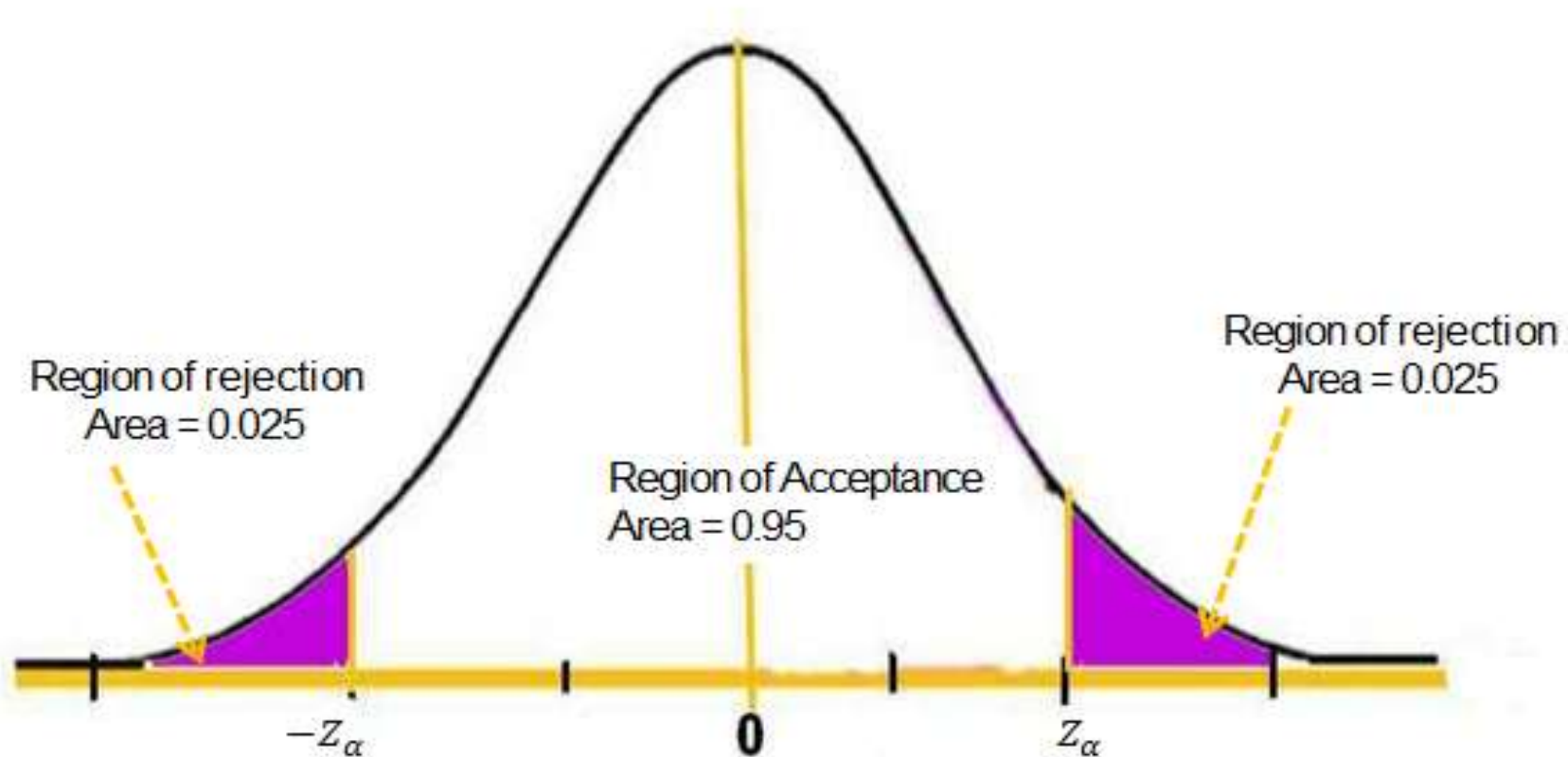


Test of Hypothesis (CO5)

Single right tailed test
if $\alpha = 5\%$



Two Tailed Test if $\alpha = 5\%$



Conclusion

If absolute calculated value of test statistic is less than or equal to its critical value(table value), we accept the null hypothesis H_0 otherwise reject it.

Test of hypothesis for large samples and Small samples

Difference between Large and Small sample

Sr. No.	Large Sample	Small Sample
1	The sample size is greater than 30.	The sample size is less than or equal to 30.
2	The value of statistic obtain from the sample can be taken as an estimate of population parameter.	The value of statistic obtain from the sample can not be taken as an estimate of population parameter.
3	Normal distribution (z-test) is used for testing.	Sampling distribution like t, F etc. are used for testing.

❖ Test of hypothesis for large samples($n > 30$)

The following are the some important applications of test of hypothesis in case of large samples-

- Test of hypothesis about population mean
- Test of hypothesis about difference between two population mean

Test of hypothesis about population mean

let $\bar{x}$ be the mean of a large random sample of size n drawn from a normal population with mean μ and S.D. σ . To test the hypothesis that population mean μ has a specified value the test statistics to be used is given by

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Test of Hypothesis (CO5)

For large sample $\sigma \cong s$

Critical value	Level of significance	
	1%	5%
Two tailed	$ Z = 2.58$	$ Z = 1.96$
One tailed	$ Z = 2.33$	$ Z = 1.64$

Example-1: A random sample of 900 members has a mean 3.4 cm. Can it be reasonably regarded as a sample from a large population of mean 3.2 cm and S.D. 2.3 cm?(Test at 5% level of significance)

Sol: Given $n = 900, \mu = 3.2, \sigma = 2.3, \bar{x} = 3.4$

$H_0: \mu = 3.2$

$H_1: \mu \neq 3.2$ (Two tailed test)

$\alpha = 0.05$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = 2.61$$

Table value $Z = 1.96$

Conclusion : $|Z|_{cal} > \text{table value}$

So H_0 is rejected.

i.e. sample not comes out from population

Example-2: An insurance agent has claimed that the avg. age of policy holders who insure through him is less than the avg. age of all agents which is 30.5 yrs.

A random sample of 100 policy holders who had insured through him gave the following age distribution-

Age	16-20	21-25	26-30	31-35	36-40
No. of person	12	22	20	30	16

Calculate arithmetic mean and standard deviation(S.D.) of this distribution and test his claim at 5% level of significance.

Test of Hypothesis (CO5)

Sol:

Age	No. of person (f)	x	$f \cdot x$	$f(x - \bar{x})^2$
16-20	12	18	216	1399.68
21-25	22	23	506	740.08
26-30	20	28	560	12.8
31-35	30	33	990	529.2
36-40	16	38	608	1354.24
Total	100		2880	4036

$$\bar{x} = \frac{\sum f_i x_i}{N} = \frac{2880}{100} = 28.8; \quad s = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}} = 6.352$$

$$H_0: \mu = 30.5$$

$$H_1: \mu < 30.5 \text{ (Single tailed test)}$$

$$\alpha = 0.05$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{x} - \mu}{s/\sqrt{n}} = -2.681$$

$$\text{Table value: } Z = 1.64$$

$$\text{Conclusion : } |Z|_{cal} > \text{table value}$$

So H_0 is rejected.

His claim is right.

❖ Test of hypothesis about difference between two population mean

Let $\bar{x}_1$ be the mean of a sample of size n_1 from a population with mean μ_1 and variance σ_1^2 . Let $\bar{x}_2$ be the mean of a sample of size n_2 from a population with mean μ_2 and variance σ_2^2 . To test whether the two population means are equal or not, the test statistic to be used as-

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Note :

- When $\sigma_1 = \sigma_2 = \sigma$ then $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

If σ is unknown and $\sigma_1 = \sigma_2$ then

$$\sigma^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$$

- If σ_1, σ_2 are unknown & $\sigma_1 \neq \sigma_2$ then

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Example: The average income of persons was Rs 210 with S.D. of Rs 10 in a sample of 100 people of a city. For another sample of 150 persons the avg. income was Rs 220 with S.D. of Rs 12. Test whether there is any significance difference between the avg. income of the localities.(Test at 5% level of significance)

Sol:

Given $n_1 = 100, n_2 = 150, \bar{x}_1 = 210,$
 $\bar{x}_2 = 220, s_1 = 10, s_2 = 12$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\alpha = 0.05$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = -7.14$$

Table value $Z = 1.96$

Conclusion : $|Z|_{cal} > \text{table value}$

So H_0 is rejected.

Yes there is a significance difference the avg. income of localities.

P-Value (Observed Significance Level):

It's the measure of the inconsistency between the hypothesized value for a population parameter and the observed sample statistic.

Using the P-Value

- Reject H_0 if $p - value \leq \alpha$
- Fail to Reject H_0 if $p - value > \alpha$

For a One-Tailed Test:

- Reject H_0 if the p -value is less than the significance level α .
- Accept H_0 otherwise.

For a Two-Tailed Test:

- Reject H_0 if the p -value is less than $\frac{1}{2}$ of the significance level α .
i.e. .025
- Accept H_0 otherwise.

Example-1 A manufacturing process produces TV. tubes with an average life $\mu = 1200$ hr and $\sigma = 300$ hrs. A new process is thought to give tubes a higher average life. And out of a sample of 100 tubes we find that they have an average life $\bar{x} = 1265$ hr. Is the new process really any better than the old?

Solution:

$$H_0: \mu = 1200$$

$$H_1: \mu > 1200 \text{ (Single tailed test)}$$

$$\alpha = 0.05$$

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = 2.17$$

$$P(\bar{x} \geq 1265) = P(z \geq 2.17) = .015$$

Here we reject H_0

Since $p - value \leq \alpha = 0.05$

Confidence Interval

We would like to determine an interval in which the population parameter is supposed to lie.

95% two sided C.L for μ is

$$\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \quad \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$$

99% two sided C.L for μ is

$$\left[\bar{x} - 2.58 \frac{\sigma}{\sqrt{n}} \quad \bar{x} + 2.58 \frac{\sigma}{\sqrt{n}} \right]$$

Example: A random sample of 200 measurements from a large population gave a mean value of 50 and S.D. of 9. Determine 95% confidence interval for mean of population.

Sol: CI for μ :

$$\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \quad \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$$
$$= [48.75 \quad 51.24]$$

Test of hypothesis for small samples ($n \leq 30$)

- t-test
- F-test

Test of hypothesis for Small Samples

When the size of the sample $n \leq 30$, then the sample is called small sample. For such sample it will not be possible for us to assume that the random sampling distribution of a statistic is approximately normal. In this sampling distribution is t-distribution.

1. Write a short note on Null hypothesis?
2. Write a short note on Alternative hypothesis?
3. A sample of 800 students from a university was taken and their average weight was found to be 112 pounds with S.D. of 20 pounds. Could the mean weight in the population be 120 pounds.(Take $\alpha = 0.05$)

- ✓ Test of hypothesis
- ✓ Procedure of Test of hypothesis
- ✓ Null Hypothesis
- ✓ Alternate Hypothesis
- ✓ Level of significance
- ✓ Test of hypothesis about difference between two population mean
- ✓ Test of hypothesis about Small samples

t-Test

- Explanation why a t distribution is associated with $n - 1$ degrees of freedom and describe the information that is conveyed by the t statistic.
- Calculation the degrees of freedom for a one-sample t test and a two-independent-sample t test, and locate critical values in the t table.
- Identify the assumptions for the one-sample t test.
- Identify the assumptions for the two-independent-sample t test.

t-distribution: It is used when sample size is ≤ 30 and the population S.D. is unknown.

Application of t-distribution:

- To test the significance (hypothesis) of the difference between the mean of small sample and mean of the population
- To test the significance of the difference between the mean of two small samples.

❖ **To test the significance of the difference between the mean of small sample and mean of the population:**

When population is distributed normally and σ is unknown then statistic

$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ follows t-distribution with $n - 1$ degree of freedom.

where $S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ = Calculated Standard deviation of sample

$\bar{x}$ → mean of sample

μ → mean of Population

n → sample size

Note-If standard deviation s of sample is given then

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

follows t-distribution with $n - 1$ degree of freedom.

$$\therefore S = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} \text{ and } s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n}}$$

From here we get

$$S = \sqrt{\frac{n}{n-1}} s$$

Example-1: The 9 items of a sample have the following values 45,47,50,52,48,47,49,53,51. Does the mean of these values differ significantly from the assumed mean 47.5.(Test at 5% level of significance)

t-Test (CO5)

Sol: Here sample is small so we use t-test

x	$(x - \bar{x})^2$
45	16.81
47	4.41
50	0.81
52	8.41
48	1.21
47	4.41
49	0.01
53	15.21
51	3.61
$\sum x = 442$	$\sum (x - \bar{x})^2 = 54.89$

$$\text{Now } S = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 2.619, \quad \bar{x} = \frac{\sum x}{n} = 49.1$$

$$H_0: \mu = 47.5$$

$$H_1: \mu \neq 47.5 \text{ (Two tailed test)}$$

$$\alpha = 0.05$$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = 1.8327$$

$$\text{Degree of freedom} = 9 - 1 = 8$$

$$\text{Table value: } t_{(8)}(0.05) = 2.306$$

$$\text{Conclusion : } |t|_{cal} < \text{table value}$$

So H_0 is accepted.

There is no significant difference between their means.

Example-2: A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units.

(Test at 5% level of significance)

Sol: here $n = 20, \bar{x} = 42, s = 5$

$H_0: \mu = 45$

$H_1: \mu \neq 45$ (Two tailed test)

$\alpha = 0.05$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = -2.61$$

Degree of freedom = $20 - 1 = 19$

Table value: $t_{(19)}(0.05) = 2.093$

Conclusion : $|t|_{cal} > \text{table value}$

So H_0 is rejected.

i.e. there is significance difference between sample mean & population mean.

Example-3: The height of 8 males participating in an athletic championship are found to be 175,168,165,170,167,160,173 and 168 cm. Can we conclude that the avg. height is greater than 165 cm.(Test at 5% level of significance)

Sol: Here $S = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 4.6521$

$$\bar{x} = \frac{\sum x}{n} = 168.25$$

$$H_0: \mu = 165$$

$$H_1: \mu > 165 \text{ (single tailed test)}$$

$$\alpha = 0.05$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = 1.97$$

$$\text{Degree of freedom} = 8 - 1 = 7$$

$$\text{Table value: } t_{(7)}(0.05) = 1.895$$

$$\text{Conclusion : } |t|_{cal} > \text{table value}$$

So H_0 is rejected.

i.e. avg. height is greater than 165.

1. A sample of 1000 students from a university was taken and their average weight was found to be 112 pounds with S.D. of 20 pounds. Could the mean weight in the population be 120 pounds.(Take $\alpha = 0.05$)
2. Write a short note on Null hypothesis?
3. Samples of size 10 and 14 were taken from two normal populations with S.D. 3.5 and 5.2. The sample means were found to be 20.3 and 18.6. Test whether the means of two populations are the same. If the tabulated value is 2.0739 at 5% level.

To test the significance of the difference between the mean of two small samples.

Let two independent sample of size n_1 and n_2 be drawn from two normal population with mean μ_1 and μ_2 and equal S.D. ($\sigma_1 = \sigma_2 = \sigma$) but unknown. To test whether the two population mean are equal or whether the difference $\bar{x}_1 - \bar{x}_2$ is significant we use t-test and the appropriate test statistic to be used is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ with } n_1 + n_2 - 2 \text{ degree of freedom.}$$

Note:

- If the samples S.D. s_1 and s_2 are given then we have

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}, \quad \text{So } S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

- If s_1 and s_2 are not given, then

$$S = \sqrt{\frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

Example-1 Samples of sizes 10 and 14 were taken from two normal populations with S.D. 3.5 and 5.2. The sample means were found to be 20.3 and 18.6. Test that whether the means of two populations are the same at 5% level.

Sol: Here $n_1 = 10, n_2 = 14, s_1 = 3.5, s_2 = 5.2, \bar{x}_1 = 20.3, \bar{x}_2 = 18.6$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\alpha = 0.05$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ where } S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} = 4.772$$

$$t = 0.8604$$

Degree of freedom = $10 + 14 - 2 = 22$, Table value: $t_{(22)}(0.05) = 2.07$

Conclusion : $|t|_{cal} < \text{table value}$, So H_0 is accepted.

Example-2: The height of 6 randomly sailors 63,65,68,69,71 and 72. Those of 9 randomly chosen soldiers are 61,62,65,66,69,70,71,72 and 73. Test whether the sailors are on the avg. taller than soldiers. (Test at 5% level of significance)

Sol: Let x_1 and x_2 be two samples denoting the height of sailors & soldiers.

Here $n_1 = 6, n_2 = 9$

$$\text{Now } S = \sqrt{\frac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}} = 4.038, \quad \bar{x}_1 = 68, \bar{x}_2 = 67.66$$

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2 \text{ (Single tailed test)}$$

$$\alpha = 0.05$$

t-Test (CO5)

x_1	x_2	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$
63	61	25	44.3556
65	62	9	32.0356
68	65	0	7.0756
69	66	1	2.7556
71	69	9	1.7956
72	70	16	5.4756
	71		11.1556
	72		18.8356
	73		28.5156
Total=408	609	60	152.0004

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ where } S = \sqrt{\frac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}} = 4.038$$

$$t = 0.1597$$

Degree of freedom = $6 + 9 - 2 = 13$

Table value: $t_{(13)}(0.05) = 1.77$

Conclusion : $|t|_{cal} < \text{table value}$

So H_0 is accepted.

i.e. the sailor are not on avg. taller than soldiers

Confidence limit for population mean

If t_α is the value of t at level of significance α at $n - 1$ degree of freedom then

For two tailed test

H_0 is accepted if $t_{cal} < \text{table value}$

$$\left| \frac{\bar{x} - \mu}{S/\sqrt{n}} \right| < t_\alpha$$

$$\Rightarrow \bar{x} - t_\alpha \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_\alpha \frac{S}{\sqrt{n}}$$

95% confidence limits(5% level of significance) are $\left[\bar{x} - t_{0.05} \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_{0.05} \frac{S}{\sqrt{n}} \right]$

99% confidence limits(1% level of significance) are $\left[\bar{x} - t_{0.01} \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_{0.01} \frac{S}{\sqrt{n}} \right]$

Example: A random sample of size 16 has 53 as mean. The sum of square of deviation from mean is 150. Find the two sided 95% confidence interval for population mean.

Sol:

$$n = 16, \bar{x} = 53, \sum (x_i - \bar{x})^2 = 150$$

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = 3.162$$

$$\text{Degree of freedom (df)} = 16 - 1 = 15$$

$$t_{15}(0.05) = 2.131$$

$$95\% \text{ confidence limits for } \mu \text{ are } \left[\bar{x} - t_{0.05} \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_{0.05} \frac{S}{\sqrt{n}} \right]$$

$$[51.315 \quad 54.685]$$

Q1: Intelligent test were given to two groups of boys & girls

	Mean	S.D.	Size
Girl	75	8	60
Boys	73	10	100

Examine if the difference between mean scores is significant? (Take $\alpha = 0.05$)

Ans: H_0 is accepted

Q2: A company wants to improve its sales. The previous sales data indicated that the average sale of 25 salesmen was \$50 per transaction. After training, the recent data showed an average sale of \$80 per transaction. If the standard deviation is \$15. Has the training provided improved the sales?

Answer: Yes, training improved the sales.

1 Intelligent test were given to two groups of boys & girls

	Mean	S.D.	Size
Girl	68	5	60
Boys	85	10	100

Examine if the difference between mean scores is significant? (Take $\alpha = 0.05$)

2. Define Null hypothesis for

- i. t test
- ii. Z-test

- ✓ Test of hypothesis
- ✓ Procedure of Test of hypothesis
- ✓ Null Hypothesis
- ✓ Alternate Hypothesis
- ✓ Level of significance
- ✓ Test of hypothesis about difference between two population mean
- ✓ t-Test

Test of hypothesis for small samples ($n \leq 30$)

- t-test
- F-test

Test of hypothesis for Small Samples

When the size of the sample $n \leq 30$, then the sample is called small sample. For such sample it will not be possible for us to assume that the random sampling distribution of a statistic is approximately normal. In this sampling distribution is F-distribution.

- **F-Test**: In testing the Significance difference of two means of two Samples, we assumed that the two Samples came from the Same Population or population with equal Variance. The object of the F-test is to discover whether two independent estimates of population variance differ significantly or whether the two samples may be regarded as drawn from the normal populations having the same variance.
- Hence before applying the t-test for the Significance of difference of two means, we have to test for the equality of population variance by using F-test.
- Let n_1 and n_2 be the Sizes of two Samples with Variances s_1^2 and s_2^2 respectively.

Case I :

- If s_1^2 & s_2^2 are given and $S_1^2 > S_2^2$. Then Test Statistic

$$F = \frac{s_1^2}{s_2^2} > 1 \text{ with d.o.f. is } \nu_1 = n_1 - 1, \nu_2 = n_2 - 1.$$

- If $S_2^2 > S_1^2$, then Test Statistic

$$F = \frac{S_2^2}{S_1^2} > 1 \text{ with d.o.f. is } \nu_2 = n_2 - 1, \nu_1 = n_1 - 1$$

$$\text{where } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} \text{ and } S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

Case II:

- If s_1^2 & s_2^2 are not given and $s_1^2 > s_2^2$. Then Test Statistic is

$$F = \frac{s_1^2}{s_2^2}, \text{ with d.o.f. is } \nu_1 = n_1 - 1, \nu_2 = n_2 - 1.$$

- If s_1^2 & s_2^2 are not given and $s_2^2 > s_1^2$. Then Test Statistic is

$$F = \frac{s_2^2}{s_1^2}, \text{ with d.o.f. is } \nu_2 = n_2 - 1, \nu_1 = n_1 - 1.$$

Where x_1 and x_2 are two Samples, $s_1^2 = \frac{\sum(x_1 - \bar{x}_1)^2}{n_1 - 1}$ and $s_2^2 = \frac{\sum(x_2 - \bar{x}_2)^2}{n_2 - 1}$.

Conclusion: Same as t-test.

Q1. The two random samples reveal the following data:

Sample no.	Size	Mean	Variance
I	16	440	40
II	25	460	42

Test the equality of Population Variance.

Ans. For Checking the equality of Population Variance, we apply F-test.

Null Hypothesis H_0 : $\sigma_1^2 = \sigma_2^2$ i.e. there is no significant difference between the population Variances.

Alternative Hypothesis H_1 : $\sigma_1^2 \neq \sigma_2^2$

Test Statistic: Since s_1^2 & s_2^2 are given then

$$\text{Hence } S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{16 \times 40}{15} = 42.7$$

$$\text{And } S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{25 \times 42}{24} = 43.75$$

Since $S_2^2 > S_1^2$, Now $F = \frac{S_2^2}{S_1^2} = \frac{43.75}{42.7} = 1.02$ with degree of freedom is $\nu_2 = n_2 - 1, \nu_1 = n_1 - 1$ i.e.

$$\nu_2 = 25 - 1 = 24 \text{ and } \nu_1 = 16 - 1 = 15 .$$

Conclusion: Here the calculate Value of $F = 1.02 < F_{(24,15)} = 2.29$ at 5% level of Significance. Hence H_0 is accepted. Therefore population variances are equal.

Example-2: The height of 6 randomly sailors 63,65,68,69,71 and 72. Those of 9 randomly chosen soldiers are 61,62,65,66,69,70,71,72 and 73. Test the equality of Population Variance. (Test at 5% LOS)

Sol: Let x_1 and x_2 be two samples denoting the height of sailors & soldiers.

Here $n_1 = 6, n_2 = 9$

Where $s_1^2 = \frac{\sum(x_1 - \bar{x}_1)^2}{n_1 - 1}$ and $s_2^2 = \frac{\sum(x_2 - \bar{x}_2)^2}{n_2 - 1}$

$\bar{x}_1 = 68, \bar{x}_2 = 67.66$

Null Hypothesis H_0 : $\sigma_1^2 = \sigma_2^2$ i.e. there is no significant difference between the population Variances.

Alternative Hypothesis H_1 : $\sigma_1^2 \neq \sigma_2^2$

Level of significance $\alpha = 0.05$

F-Test (CO5)

x_1	x_2	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$
63	61	25	44.3556
65	62	9	32.0356
68	65	0	7.0756
69	66	1	2.7556
71	69	9	1.7956
72	70	16	5.4756
	71		11.1556
	72		18.8356
	73		28.5156
Total=408	609	60	152.0004

$$s_1^2 = \frac{\sum(x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{60}{5} = 12 \text{ and}$$

$$s_2^2 = \frac{\sum(x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{152.0004}{8} = 19.00005$$

$$\text{Test Statistic F test} = \frac{s_2^2}{s_1^2} = \frac{19.00005}{12} = 1.583375$$

if $s_2^2 > s_1^2$ with degree of freedom is

$$\nu_2 = n_2 - 1 = 9 - 1 = 8, \nu_1 = n_1 - 1 = 6 - 1 = 5.$$

Table value: $F_{(8,5)}(0.05) = 4.82$

Conclusion : $|F|_{cal} < \text{table value}$

So H_0 is accepted.

i.e. Therefore population variances are equal.

Q1: The random samples were drawn from two normal populations and the following results were obtained

Sample 1: 16,17,18,19,20,21,22,24,26,27

Sample 2: 19,22,23,25,26,28,29,30,31,32,35,36

Obtain estimates of the variances of populations and test whether the two populations have the same variances.

(Take $\alpha = 0.05$)

Ans:

Q2: A toy manufacturer wants to get batteries for toys. A team collected 41 samples from supplier A and the variance was 110 hours. The team also collected 21 samples from supplier B with a variance of 65 hours. At a 0.05 alpha level determine if there is a difference in the variances.

Ans: H_0 is rejected.

Chi Square Test(χ^2)

- Distinguish between nonparametric and parametric tests.
- Explanation how the test statistic is computed for a chi-square test.
- Compute the chi-square goodness-of-fit test and interpret the results.
- Identify the assumption and the restriction of expected frequency size for the chi-square test.
- Compute the chi-square test for independence and interpret the results.

Chi Square Test(χ^2)

1. Chi-square test of goodness of Fit.
2. Chi-square test of independence of Attributes.

Chi Square Test(χ^2)

- Chi Square test is the test of significance.
- Chi square test is a useful measures of comparing experimentally obtained result with those expected theoretically and based on hypothesis.
- $$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$
- It is a mathematical expression, representing the ratio of experimentally obtained result(O) and theoretically expected result(E) based on certain hypothesis. It uses data in form of frequencies.

Chi Square Test(χ^2)(CO5)

- Chi square test is calculated by dividing the square of overall deviation in the observed and expected frequencies by expected frequency. $\chi^2 = \sum \frac{(O-E)^2}{E}$

O=the frequencies observed, E=the frequencies expected, Σ = sum of

- If there is no difference between actual and observed frequencies, the value of Chi- Square is zero.
- If there is difference between actual and observed frequencies, the value of Chi- Square would be more than zero.
- But the difference in the observed frequencies, may also be due to the sampling fluctuations and it should be ignored in drawing inference.

Degree of Freedom

- In test, while comparing the calculated value with the table value, we have to calculate the degree of freedom. The degree of freedom is calculated from the no of classes. Therefore, the no. of degree of freedom in a test is equal to the no. of classes minus one.
- If there are two classes, three classes and four classes, the degree of freedom would be 2-1, 3-1 and 4-1, respectively. In a contingency table, the degree of freedom is calculated in different manner:

$$d.f. = (r-1)(c-1)$$

Where r = no. of row in the table, C = no. of column in the table

- Thus in 2×2 contingency table, the degree of freedom is $(2-1)(2-1)=1$. Similarly, in a 3×3 contingency table, the degree of freedom is $(3-1)(3-1)=4$.

Chi Square Test(χ^2)(CO5)

Likewise in 3×4 contingency table degree of freedom is $(3-1)(4-1)=6$, and so on.

Example: 2 Rows $\times$ 2 columns

	COLUMN 1	COLUMN 2	ROW TOTAL
ROW 1	+	+	RT 1
ROW 2	+	+	RT 2
COLUMN TOTAL	CT 1	CT 2	

Degree of freedom = $(r-1)(c-1)=(2-1)(2-1)=1$

Application of chi square test

- The chi square are applicable in varied problems in agriculture, biology engineering and medical sciences.
- To test the goodness of fit.
- To test the independence of attributes.
- To test the detection of linkage.
- To test the homogeneity of independent estimates of the population variance.

Chi Square Test(χ^2)(CO5)

Example: The demand for a particular spare part in a factory was found to vary from day-to-day. In a sample study the following information was obtained:

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
No. of parts demanded	1124	1125	1110	1120	1126	1115

Under the null hypothesis, the expected frequencies of the spare part demanded on each of the six days would

$$\text{be: } (1124 + 1125 + 1110 + 1120 + 1126 + 1115) / 6 = 6270 / 6 = 1120$$

Chi Square Test(χ^2)(CO5)

Days	Observed (O_i)	Expected (E_i)	($O_i - E_i$)	($O_i - E_i$) ²
Monday	1124	1120	4	16
Tuesday	1125	1120	5	25
Wednesday	1110	1120	-10	100
Thursday	1120	1120	0	0
Friday	1126	1120	6	36
Saturday	1115	1120	-5	25
Total	6720	6720		

Chi Square Test(χ^2)(CO5)

$$\chi^2 = \sum_{i=1}^n \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} = 0.179$$

The number of degrees of freedom = $6 - 1 = 5$ (since we are given 6 frequencies subjected to only one linear constraint: $\sum O_i = \sum E_i = 6720$).
The tabulated $\chi^2_{0.05}$ for 5 d.f. = 11.07

Conclusion: Since calculated value of χ^2 is less than the tabulated value, it is not significant and the null hypothesis may be accepted at 5% level of significance. Hence we conclude that the numbers of parts demanded are same over the 6-day period.

Chi square test of Independent of Attributes:

- Let us consider two attributes A and B divides into r classes $A_1, A_2, A_3 \dots A_r$ and B divided into s classes $B_1, B_2, B_3 \dots B_s$.
- Such a classification in which attributes are divided into more than two classes is known as manifold classification.

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^s \frac{[(A_i B_j) - (A_i B_j)_o]^2}{(A_i B_j)_o} = \sum_{i=1}^r \sum_{j=1}^s \frac{[O_{ij} - E_{ij}]^2}{E_{ij}}$$

Where $O_{ij}=A_iB_j$: observed frequency for contingency table category in column i and row j.

$E_{ij}=(A_iB_j)_o$: expected frequency for contingency table category in column i and row j.

$$(A_iB_j)_o = \frac{(A)_i(B)_j}{N}, (i=1,2,3\dots r, j=1,2,3\dots s)$$

Which is distributed as a χ^2 with $(r-1)(s-1)$ d.f.

Example :Two sample polls of votes for two candidates A and B for public office are taken, one from among the resident area of ruler areas. The results are given in the following table. Examine whether the nature of area related to the voting preference in elections.

Chi Square Test(χ^2)(CO5)

Area	Voter for A	Voter for B	Total
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

Solution : H_0 : the nature of the area is independent of the voting preference in the election

Expected frequencies

$$E(620) = \frac{1170 \times 1000}{2000} = 585$$

$$E(380) = \frac{830 \times 1000}{2000} = 415$$

$$E(550) = \frac{1170 \times 1000}{2000} = 585$$

$$E(450) = \frac{830 \times 1000}{2000} = 415$$

Calculate the statistic χ^2

$$= \frac{(620-585)^2}{585} + \frac{(380-415)^2}{415} + \frac{(550-585)^2}{585} + \frac{(450-415)^2}{415}$$

$$= 10.0891$$

The tabulated $\chi^2_{0.05}$ for (2-1)(2-1) d.f. = 3.841

Conclusion: since calculate χ^2 is much greater than the tabulated value, it is highly significant and null hypothesis is rejected at 5% of level of significance. Thus we conclude that nature of area is rejected to voting preference in the election.

Chi Square Test(χ^2)(CO5)

Example : What are the expected frequencies of the contingency tables given below:

Observed frequency:

p	q	$p + q$
r	s	$r + s$
$p + r$	$q + s$	$p + q + r + s$



Expected frequency:

$\frac{(p + r)(p + q)}{(p + q + r + s)}$	$\frac{(q + s)(p + q)}{(p + q + r + s)}$
$\frac{(p + r)(r + s)}{(p + q + r + s)}$	$\frac{(q + s)(r + s)}{(p + q + r + s)}$

Chi Square Test(χ^2)(CO5)

Example : To test the effectiveness of inoculation against cholera, the following table was obtained:

	Attacked	Not Attacked	Total
Inoculated	30	160	190
Not Inoculated	140	460	600
Total	170	620	790

(figures represent the number of persons.)

Use χ^2 test to defend or refute the statement that the inoculation prevents attack from cholera.

Solution : H_0 :the inoculation does not prevents attack from cholera.

Chi Square Test(χ^2)(CO5)

Expected frequencies

O_i	30	160	140	690
E_i	40.886	149.11	129.11	470.89
$\frac{(O_i - E_i)^2}{E_i}$	2.898	0.795	0.918	0.252

Calculate the statistic χ^2

	Attacked	Not Attacked
Inoculated	$\frac{190 \times 170}{790} = 40.886$	$\frac{190 \times 620}{790} = 149.11$
Not Inoculated	$\frac{600 \times 170}{790} = 129.11$	$\frac{600 \times 620}{790} = 470.89$

$$\chi^2_{cal} = \sum \frac{(O_i - E_i)^2}{E_i} = 4.8863$$

The tabulated $\chi^2_{0.05}$ for (2-1)(2-1) d.f. = 3.841

Conclusion: since calculate χ^2 is much greater than the tabulated value, it is highly significant and null hypothesis is rejected at 5% of level of significance. Hence we defend the statement that inoculation prevents attack from cholera.

Q1: A sample analysis of examination results of 500 students it was found that 220 students have failed, 170 have secured a third class, 90 have secured a second class and the rest, a first class. Do these figures support the general belief that above categories are in the ratio 4:3:2:1 respectively?(Test at 5 % level of significance)

Ans: H_0 rejected.

Q2: A sample of 400 students of under graduate and 400 students of post graduate classes was taken to know their opinion about autonomous colleges. 290 of the under graduate and 310 of the post graduate students favored the autonomous status. Present these facts in the form of a table and test at 5% level that the opinion regarding autonomous status of colleges is independent of the level of classes of students.

Ans: H_0 rejected.

- ✓ Test of hypothesis
- ✓ Test of hypothesis about difference between two population mean
- ✓ To test the significance of the difference between the mean of two small samples.
- ✓ χ^2 test

Analysis of variance(ANOVA)

- Appropriately interpret results of **analysis of variance** tests.
- Distinguish between one and two factor **analysis of variance** tests.
- Identify the appropriate hypothesis testing procedure based on type of outcome variable and number of samples.

ANOVA:

- ANOVA is used to determine there is any significant difference between the mean of three or more independent samples.
- ANOVA is carried out on the basis of ratio between two variances, this ratio forms the test statistic known as F -statistics given by-

$$F = \frac{\text{variance between the samples}}{\text{variance within the sample}}$$

ANOVA is to split up the total variation as follows:

1. Variance between samples.
2. Variance within samples.

ANOVA classification

1. One way ANOVA: This is used to see the effect of an independent variable on a given dependent variable.
2. Two way ANOVA: This is used to see the effect of two independent variable on a given dependent variable.

Analysis of Variance (ANOVA)(CO5)

ANOVA TABLE:

Source of Variation	Sum of Squared	Degree of Freedom	Mean Sum of Squares	Variance Ratio OR F
Between Samples (Column)	SSC	$c - 1$	$MSC = \frac{SSC}{c - 1}$	$F = \frac{MSC}{MSE}$
With in Samples (Error)	SSE	$n - c$	$MSE = \frac{SSE}{n - c}$	
Total	SST	$n - 1$		

Computation of test statistic

$$F = \frac{\text{variance between the samples}}{\text{variance within the samples}} = \frac{MSC}{MSE}$$

Analysis of Variance (ANOVA)(CO5)

Q.1 It Is desired to compare three Hospitals with regards to number of deaths per month. A sample of death records of each hospital and the number of deaths was given below. From these data suggest a difference in the no. of the deaths per month among three hospitals.

A	B	C7
3	6	7
4	3	3
3	3	4
5	4	6
0	4	5

Sol: Null hypothesis H_0 : There is no difference in the no. of deaths per months among three hospitals.

Alternate hypothesis H_1 : There is a significant difference in the no. of death per month among three hospitals.

Level of Significance : We use 5% level of Significance.

Test Statistic : To find the Variance ratio F , we set up an ANOVA tables as follows:

Sample totals:

$$\sum y_a = 3 + 4 + 3 + 5 + 0 = 15$$

$$\sum y_b = 6 + 3 + 3 + 4 + 4 = 20$$

$$\sum y_c = 7 + 3 + 4 + 6 + 5 = 25$$

1. Grand total(T)= $\sum y_a + \sum y_b + \sum y_c = 60$
2. Correction factor(C.F.)= $\frac{(T)^2}{n} = \frac{60^2}{15} = 240$

Sum of Squares of Samples:

$$\sum y^2_A = 3^2 + 4^2 + 3^2 + 5^2 + 0^2 = 59$$

$$\sum y^2_B = 6^2 + 3^2 + 3^2 + 4^2 + 4^2 = 86$$

$$\sum y^2_C = 7^2 + 3^2 + 4^2 + 6^2 + 5^2 = 135$$

$$\begin{aligned}\text{Total Sum of Squares} &= \sum y^2_A + \sum y^2_B + \sum y^2_C - \text{C.F.} \\ &= 59 + 86 + 135 - 240 = 40\end{aligned}$$

Sum of Squares between Samples

$$= \frac{(\sum y_A)^2}{n_1} + \frac{(\sum y_B)^2}{n_2} + \frac{(\sum y_C)^2}{n_3} - \frac{(T)^2}{n}$$

$$= \frac{(15)^2}{5} + \frac{(20)^2}{5} + \frac{(25)^2}{5} - 240$$

$$= 10$$

Sum of squares with in the samples

= Total sum of squares – sum of squares between the samples

$$= 40 - 10$$

$$= 30$$

Analysis of Variance (ANOVA)(CO5)

Degree of freedom for total sum of squares $=n-1=15-1=14$

Degree of freedom for Hospitals $=c-1 = 3-1=2$

Degree of freedom for Error $=n-c =15-3=12$

ANOVA TABLE

Source of Variation	Sum of Squared	Degree of Freedom	Mean Sum of Squares	Variance Ratio OR F
Between Samples (Column)	10 (SSC)	2	$MSC = \frac{SSC}{c-1} = \frac{10}{2} = 5$	$F = \frac{MSC}{MSE}$ $= \frac{5}{2.5} = 2$
With in Samples (Error)	30 (SSE)	12	$MSE = \frac{SSE}{n-c} = \frac{30}{12}$ $= 2.5$	
Total	40 (SST)	14		

The tabular value of F at 5 % level of significance with degree of freedom $\nu_1=2$ & $\nu_2 = 12$ is 3.89:

Conclusion: $F_{cal} < F_{tab}$, i. e. H_0 is accepted, so the difference is insignificant and we conclude that the data do not suggest a difference in the no of deaths per month among the three hospitals.

Q.2 A manufacturing company purchased three new machines of different makes and wishes to determine whether one of them is faster than the others in producing a certain out put . Five hourly production figures are observed at random from each machine and the results are given below:

Observations	A1	A2	A3
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	25
5	31	35	28

Use ANOVA and determine whether the machines are significantly Different in their mean speed.(Given at 5% level, $F_{2,12}=3.89$)

Sol. Null hypothesis H_0 : Machines are not significantly different in their mean speed i.e., $\mu_1 = \mu_2 = \mu_3$.

Alternate hypothesis H_1 :Machines are significantly different in their mean speed.

Level of Significance: We use 5% level of Significance.

Test Statistic: To find the Variance ratio F , we set up an ANOVA tables as follows: let us shift the origin at 30 i.e., reduce each observations by 30.we have

Analysis of Variance (ANOVA)(CO5)

Sample totals:

$$\sum A_1 = -5 + 0 + 6 + 8 + 1 = 10$$

$$\sum A_2 = 1 + 9 + 8 + 12 + 5 = 35$$

$$\sum A_3 = -6 + 0 - 2 - 5 - 2 = -15$$

Observations	A1	A2	A3
1	-5	1	-6
2	0	9	0
3	6	8	-2
4	8	12	-5
5	1	5	-2

1. Grand total(T)= $\sum A_1 + \sum A_2 + \sum A_3 = 30$
2. Correction factor(C.F.)= $\frac{(T)^2}{n} = \frac{(30)^2}{15} = 60$

Sum of Squares of Samples:

$$\sum A^2_1 = (-5)^2 + 0 + 6^2 + 8^2 + 1^2 = 126$$

$$\sum A^2_2 = 1^2 + 9^2 + 8^2 + 12^2 + 5^2 = 315$$

$$\sum A^2_3 = (-6)^2 + 0^2 + (-2)^2 + (-5)^2 + (-2)^2 = 69$$

3. Total Sum of Squares= $\sum A^2_1 + \sum A^2_2 + \sum A^2_3 - \text{C.F.}$
 $= 126 + 315 + 69 - 60 = 450$

Sum of Squares between Samples

$$\begin{aligned} &= \frac{(\sum A_1)^2}{n_1} + \frac{(\sum A_2)^2}{n_2} + \frac{(\sum A_3)^2}{n_3} - \frac{(T)^2}{n} \\ &= \frac{(10)^2}{5} + \frac{(35)^2}{5} + \frac{(-15)^2}{5} - 60 \\ &= 250 \end{aligned}$$

Sum of squares within the samples

$$\begin{aligned} &= \text{Total sum of squares} - \text{sum of squares between the samples} \\ &= 450 - 250 \\ &= 200 \end{aligned}$$

Degree of freedom for total sum of squares $= n - 1 = 15 - 1 = 14$

Degree of freedom for Machines $= c - 1 = 3 - 1 = 2$

Analysis of Variance (ANOVA)(CO5)

Degree of freedom for Error = $n - c = 15 - 3 = 12$

ANOVA TABLE

Source of Variation	Sum of Squared	Degree of Freedom	Mean Sum of Squares	Variance Ratio OR F
Between Samples (Column)	250 (SSC)	2	$MSC = \frac{SSC}{c - 1} = \frac{250}{2}$ $= 125$	$F = \frac{MSC}{MSE}$ $= \frac{125}{16.67}$ $= 7.498$
With in Samples (Error)	200 (SSE)	12	$MSE = \frac{SSE}{n - c} = \frac{200}{12}$ $= 16.67$	
Total	450 (SST)	14		

The tabular value of F at 5 % level of significance with degree of freedom $\nu_1=2$ & $\nu_2 = 12$ is 3.89(given)

Conclusion: $F_{cal} > F_{tab}$, so the null hypothesis is rejected and the difference is significant and we conclude that there is significant difference in the mean speed of the machines.

- ✓ Test of hypothesis
- ✓ Test of hypothesis about difference between two population mean
- ✓ To test the significance of the difference between the mean of two small samples.
- ✓ χ^2 test
- ✓ Anova

Control Charts

- That process variable can be plotted on a **control chart** over time. The **objective** of the **control chart** is to find any "special" causes of variation as well as to reflect the process improvements that have been made.

Statistical Quality Control (SQC)

SQC refers to the use of statistical methods to monitoring & maintaining the quality of product & services.

Method of SQC:

SQC are applied to two distinct phase of plant operation, they are

1. Process control
2. Product control

1. Process control:

- Under the process control, the quality of the product is controlled while the product are in process of production. It is based on prob. Theoretically, It is common that when several identical parts are manufactured some are little large & some are little small but most will be of same size when the frequency or count of the items by size is plotted with size on the horizontal scale and count on the vertical scale a normal curve is obtained. The process control is secured with the technique of **control charts**.

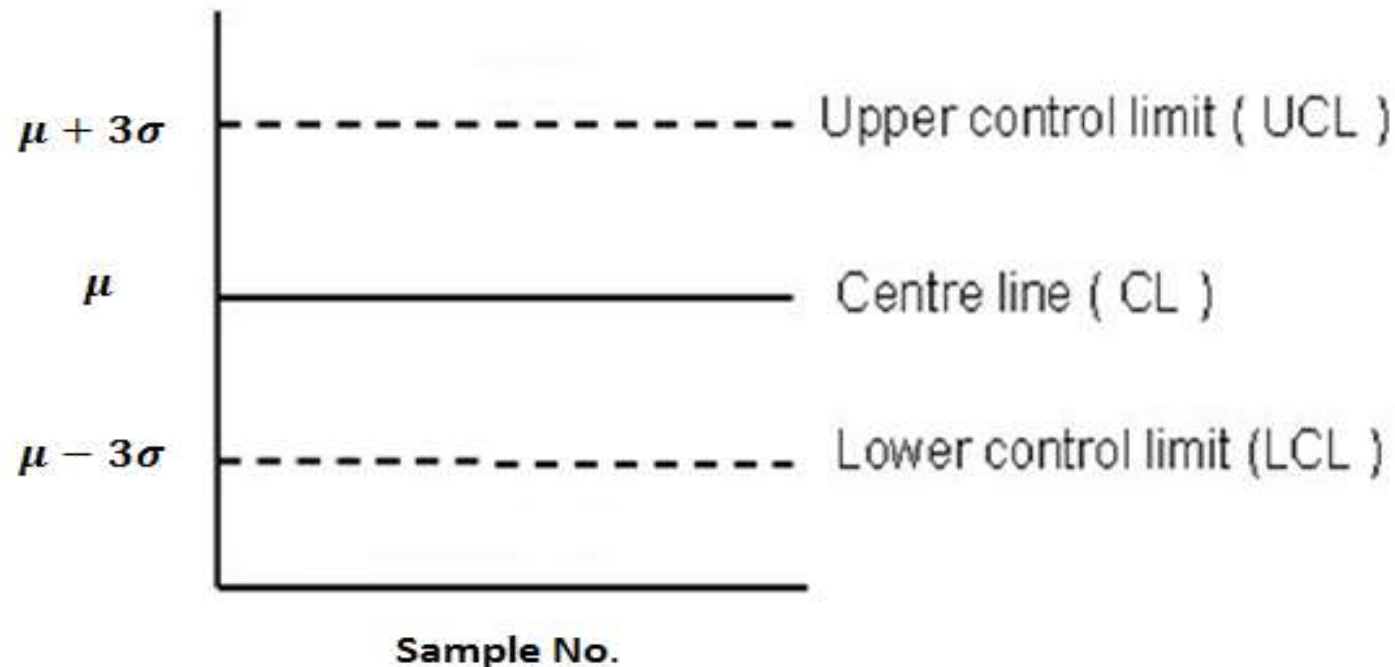
2. Product control:

- Under the product control, the quality of the product is controlled while the product is ready for sale and dispatch to the customers.
- The product control is secured with the technique of acceptance sampling. Acceptance sampling is based on premises that a sample represent the whole lot from which the sample is drawn.
- In this method samples are taken out & are carefully inspected to detect defects. On the basis of defect found, the lot is accepted or rejected. If defect are few, lot is accepted. It is rejected when defect are more. Thus acceptance sampling is used to take decision regarding acceptance or rejection a lot without examine the entire lot.

Control charts: Control chart is a graphical chart for detecting unnatural pattern of variation in the production process and determining the permissible limits of variation. There are 3 horizontal line in control chart-

1. **Control line(CL):** it passes through the middle of the chart and parallel is to the base. It represent the prescribed quality of product.
2. **Upper Control limit(UCL):** It is shown in the chart by a dotted line that passes through the chart above and parallel to the central line and represent upper limit of tolerance.
3. **Lower Control limit(LCL):** It is shown in the chart by a dotted line that passes through the chart below and parallel to the central line and represent lower limit of tolerance.

Control Charts (CO5)



Note:

- If LCL is negative then we take $LCL=0$.
- Control chart are based on 3 sigma limits.

Control chart for variables- are used to monitor characteristic that can be measured for e.g- length, weight, diameter, time etc. such charts are of two types-

1. Mean chart ($\bar{x}$ chart)
2. Range chart (R chart)

1. **Mean chart ($\bar{x}$ chart):** It is used to monitor the change in mean of a process.

Construction of $\bar{x}$ chart:

- Compute the mean of each sample.
i.e. $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$ (here k is no. of sample)
- Compute the mean of sample mean.
i.e. $\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_k}{k}$
- Compute the range:
 $R = (\text{largest value} - \text{smallest value})$ of
each sample i.e. $R_1, R_2, \dots, R_k$.

- Compute the mean of sample range

$$\text{i.e. } \bar{R} = \frac{R_1 + R_2 + \dots + R_k}{k}$$

- $CL = \bar{\bar{x}}$

$$UCL = \bar{\bar{x}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{R}$$

where A_2 are given in the table and depends upon the sample size.

- If the sample mean of each sample lie with in the UCL and LCL then process is in control. If it lie outside then process is out of control.

2. Range chart(R-chart)- It is used to monitor the dispersion or variability of the process.

Construction of R-chart-

- Compute the range:

$R = (\text{largest value} - \text{smallest value})$ of each sample i.e. $R_1, R_2, \dots, R_k$.

- Compute the mean of sample range

$$\text{i.e. } \bar{R} = \frac{R_1 + R_2 + \dots + R_k}{k}$$

- $CL = \bar{R}$

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

where D_4, D_3 are constant whose value depends upon the sample size and coming from table.

- If the range of each sample lie with in UCL and LCL then process is in control. If it lie outside then process is out of control.

Example-1: Construct mean chart ($\bar{x}$ chart) and range chart(R-chart) for the following data of 5 samples with each set of 5 items-(conversion factor for $n = 5, A_2 = 0.577, D_3 = 0, D_4 = 2.115$)

Control Charts (CO5)

Sample No.	Weights				
1	20	15	10	11	14
2	12	18	10	8	22
3	21	19	17	10	13
4	15	12	19	14	20
5	20	19	26	12	23

Control Charts (CO5)

Sol:

Sample No.	Weights					Sample mean	Range $R = L - S$
1	20	15	10	11	14	14	10
2	12	18	10	8	22	14	14
3	21	19	17	10	13	16	11
4	15	12	19	14	20	16	8
5	20	19	26	12	23	20	14
Total						80	57

$\bar{x}$ chart:

$$\bar{x}_1 = \frac{70}{5} = 14 \text{ similarly } \bar{x}_2 = 14, \bar{x}_3 = 16, \bar{x}_4 = 16, \bar{x}_5 = 20$$

$$R_1 = \text{largest} - \text{smallest} = 20 - 10 = 10$$

$$R_2 = 22 - 8 = 14$$

$$R_3 = 11$$

$$R_4 = 8$$

$$R_5 = 14$$

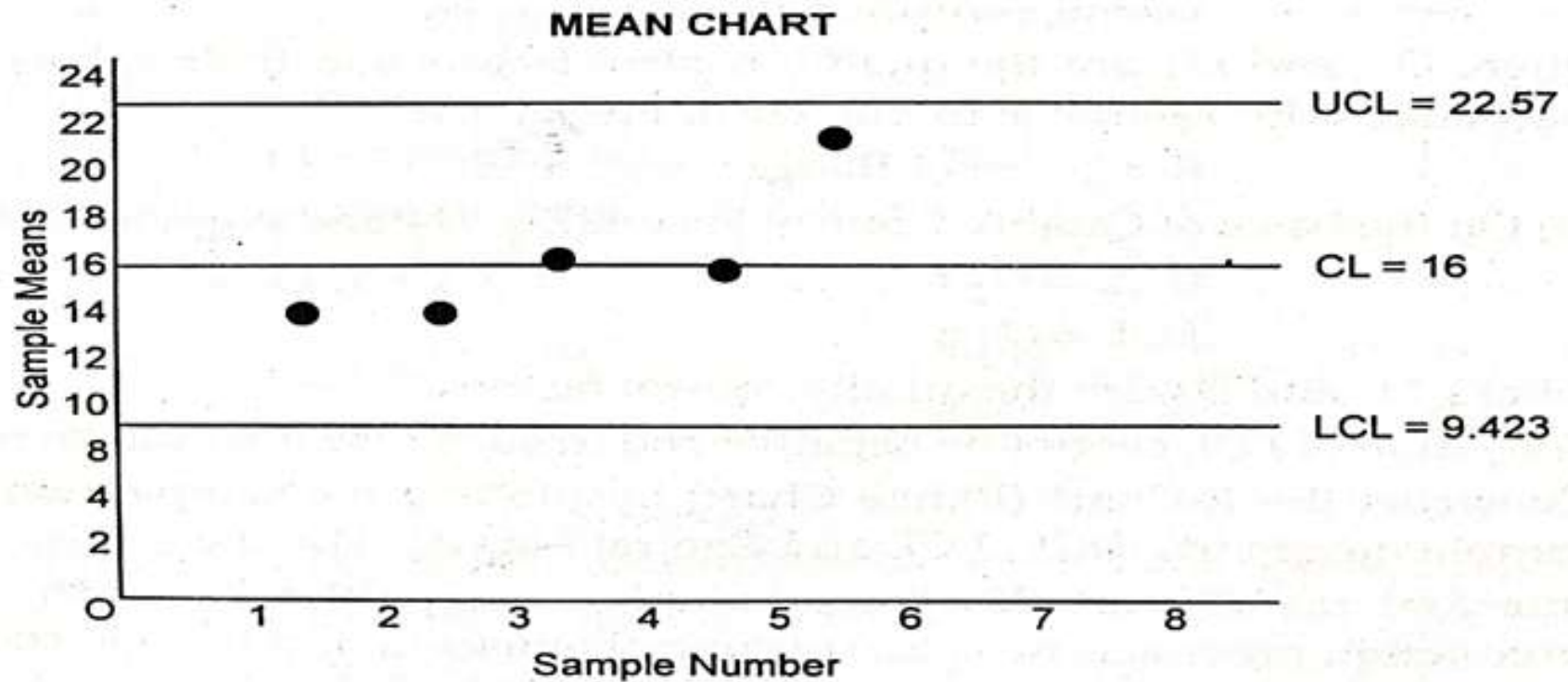
$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_5}{5} = \frac{80}{5} = 16$$

$$\bar{R} = \frac{R_1 + R_2 + \dots + R_5}{5} = \frac{57}{5} = 11.4$$

- $CL = \bar{\bar{x}} = 16$

$$UCL = \bar{\bar{x}} + A_2 \bar{R} = 22.577 \text{ here } A_2 = 0.577 \text{ (Given) } LCL = \bar{\bar{x}} - A_2 \bar{R} = 9.423$$

Mean Chart-

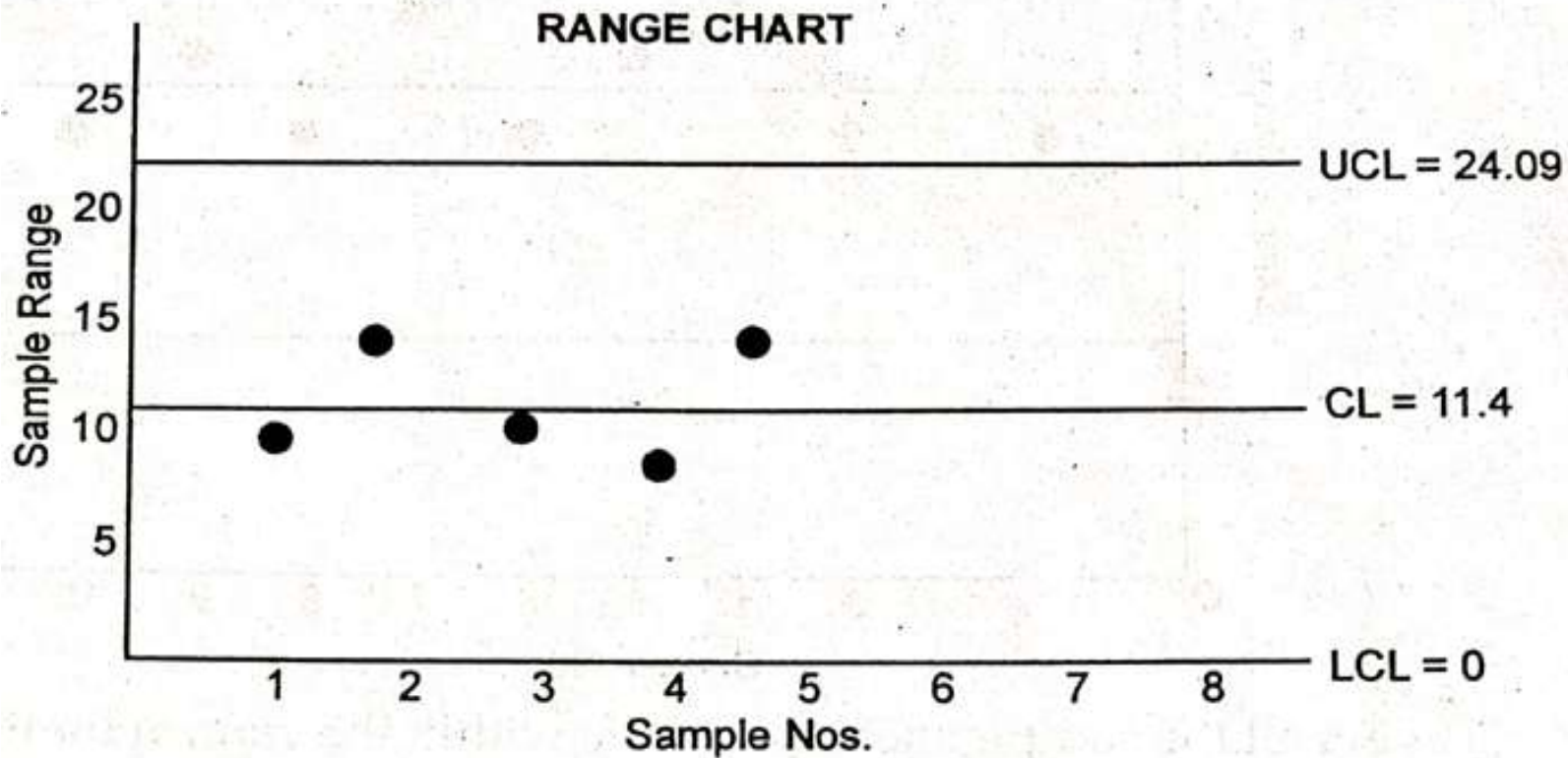


As all the sample mean values fall within the control limits, the chart shows that the given process is in statistical control.

R-chart:

- $CL = \bar{R} = 11.4$
 $UCL = D_4 \bar{R} = 24.09 \quad (\because D_4 = 2.115)$
 $LCL = D_3 \bar{R} = 0 \quad (\because D_3 = 0)$

Control Charts (CO5)



As all the range points fall within the control limits, so R -chart shows that the given process is in statistical control.

Q1. An inspection of 10 samples of size 100 each from 10 lots reveal the following number of defectives: 16,18,11,18,21,10,20,18,17,21. Do these indicate that the quality characteristics inspected is under statistical control.

Ans. Process is in under statistical control

Q2. A machine drills hole in a pipe with a mean diameter of 0.532 cm and a standard deviation of 0.002 cm. Calculate the control limits for mean of samples 5.

Ans. $CL=0.532$, $UCL=0.5346$, $LCL=0.5293$

Control chart for attributes: are used to monitor characteristic that have discrete values & can be counted. e.g- percentage of defective etc. such charts are of three types-

1. p -chart(fraction/proportion defective chart)
2. np -chart(no. of defective chart)/d-chart
3. C-chart(no. of defect per Module chart)

***p*-chart:** is designed to control the percentage or proportion/fraction of defective per samples.

Construction of *p*-chart:

- Calculate fraction defective *p* for each sample

i.e. Fraction defective $p = \frac{\text{no.of defective item}}{\text{size of the sample}}$

- Calculate avg. fraction defective ($\bar{p}$)

i.e. $\bar{p} = \frac{\text{Total no.of defective}}{\text{Total no.of unit inspected}}$

- $CL = \bar{p}$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

where $n = \frac{\text{total no. of unit inspected}}{\text{no. of sample}}$

- If the fraction defective of each sample lie within the UCL and LCL then process is in control. If it lies outside then process is out of control.

Example-1 The following data refers to visual defects found during the inspection of the first 10 samples of size 100 each from a lot of two wheelers manufactured by automobile company:

No. of samples (each of 100 items)	1	2	3	4	5	6	7	8	9	10
No. of defectives	5	3	3	6	5	6	8	10	10	4

Construct a control chart for fraction defective. What conclusions you draw from the control chart?

Control Charts (CO5)

Sol: Here no of sample $k = 10$

No. of samples	No. of units in sample	No. of defective (d)	Fraction defective (p)
1	100	5	0.05
2	100	3	0.05
3	100	3	0.03
4	100	6	0.06
5	100	5	0.05
6	100	6	0.06
7	100	8	0.08
8	100	10	0.1
9	100	10	0.1
10	100	4	0.04
Total	1000	60	0.60

$$\text{Avg. fraction defective } \bar{p} = \frac{\text{Total no.of defective}}{\text{Total no.of unit inspected}} = \frac{60}{10 \times 100} = 0.06$$

$$n = \frac{\text{total no.of unit inspected}}{\text{no.of sample}} = \frac{10 \times 100}{10} = 100$$

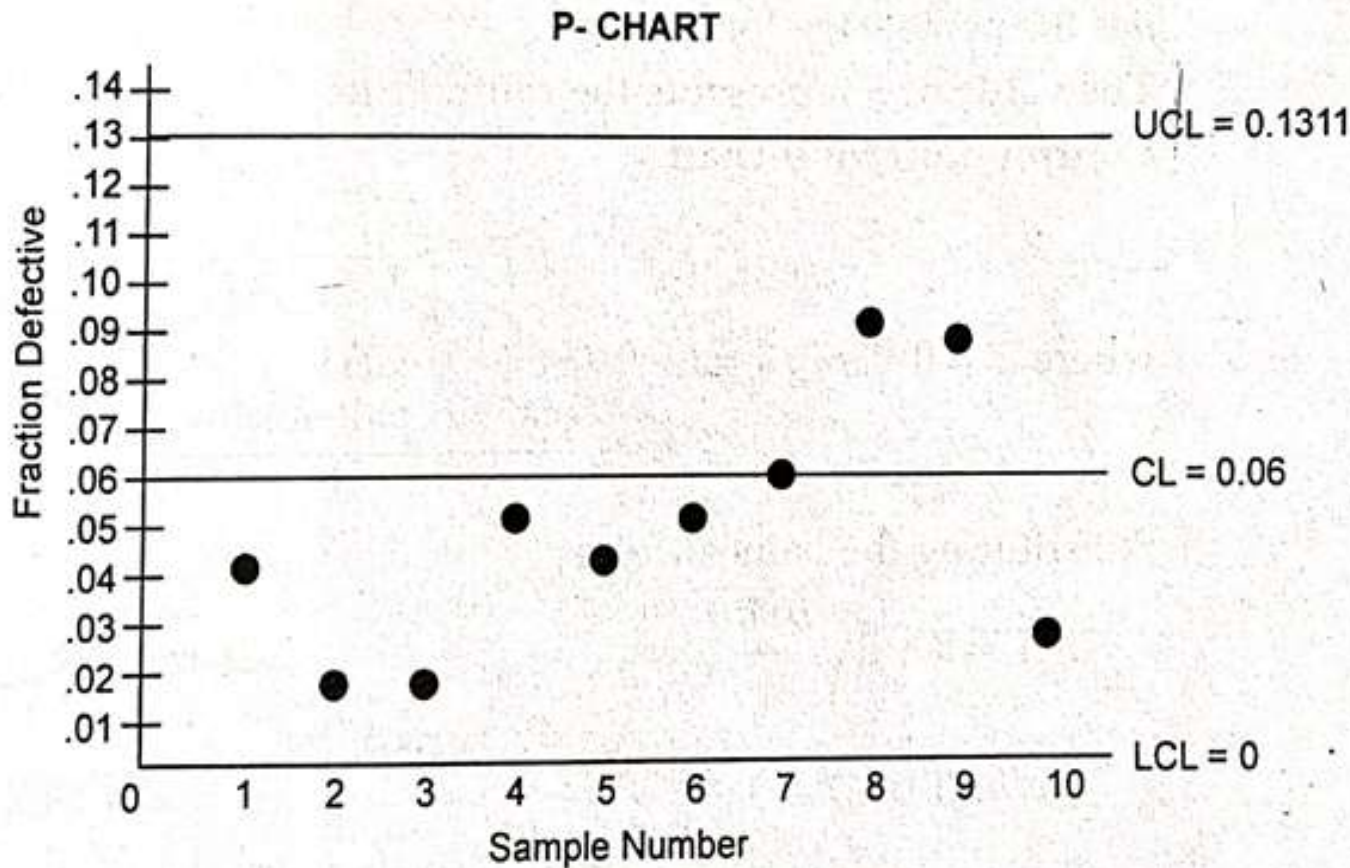
$$\text{Now CL} = \bar{p} = 0.06$$

$$\text{UCL} = \bar{p} + 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1311$$

$$\text{LCL} = \bar{p} - 3 \sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = -0.0112$$

Here LCL is negative so we take LCL=0.

Control Charts (CO5)



The above chart shows that all the points lie within the control limits. This suggests that the process is in control.

np -chart: when the sample size remain constant, np - chart is used to control the actual no. of defective per sample. The construction of np -chart is similar to p - chart because in this chart we can directly plot the no. of defective rather than fraction of defective.

Construction of np -chart:

- Calculate avg. fraction defective ($\bar{p}$)

$$\text{i.e. } \bar{p} = \frac{\text{Total no.of defective}}{\text{Total no.of unit inspected}}$$

$$CL = n\bar{p}, \quad UCL = n\bar{p} + 3\sqrt{n\bar{p}(1 - \bar{p})}, \quad LCL = n\bar{p} - 3\sqrt{n\bar{p}(1 - \bar{p})}$$

$$\text{where } n = \frac{\text{total no.of unit inspected}}{\text{no.of sample}}$$

If the no. of defectives of each sample lie with in the UCL and LCL then process is in control. If it lie outside then process is out of control.

Example-1: An inspection of 10 samples of size 400 each from 10 lots reveal the following number of defectives: 17,15,14,26,9,4,19,12,9,15
Calculate the control limits for the number of defective units. Plot on the graph and state whether the process is under control or not.

Sol: Here no. of samples= 10

Size of sample:

$$n = \frac{\text{total no. of unit inspected}}{\text{no. of sample}} = \frac{10 \times 400}{10} = 400$$

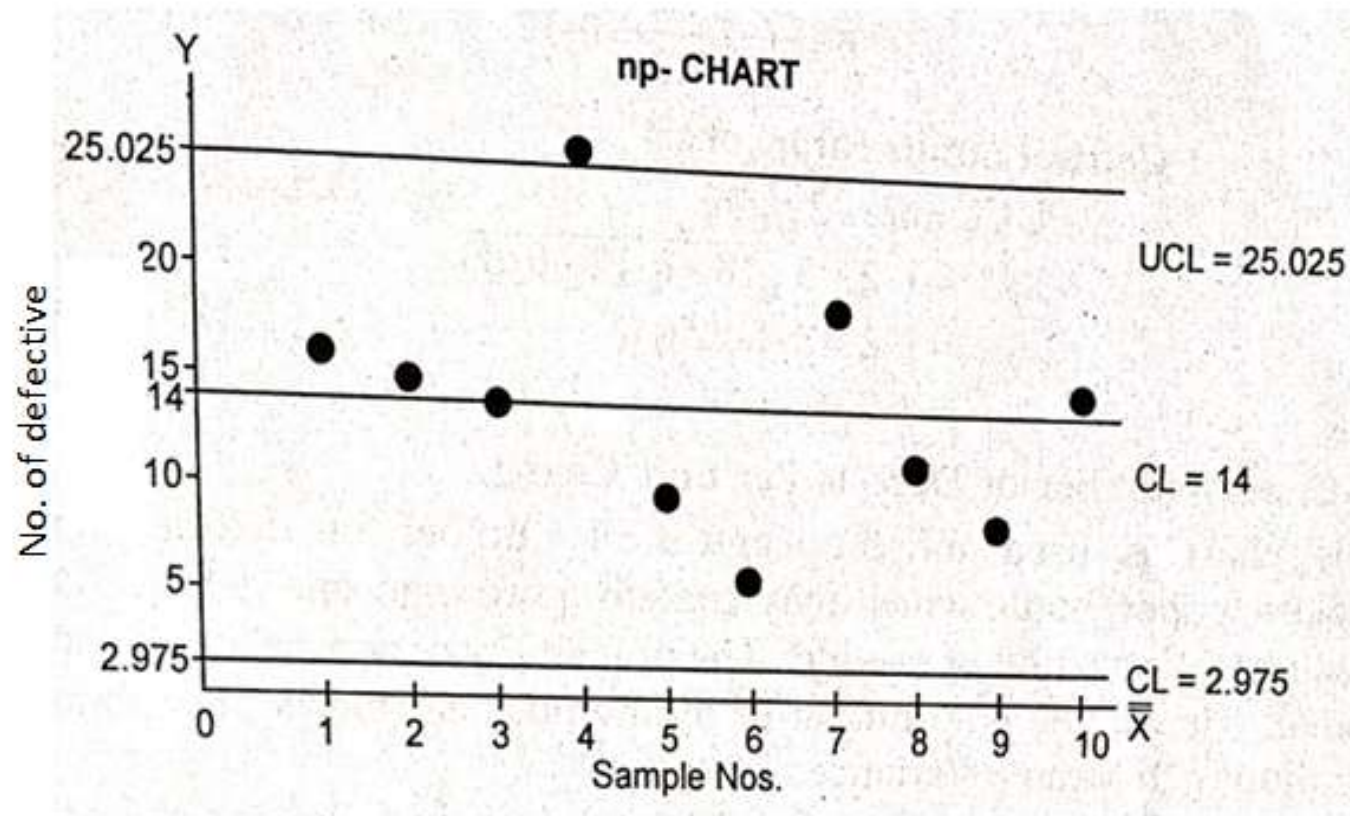
$$\text{Avg. fraction defective } \bar{p} = \frac{\text{Total no. of defective}}{\text{Total no. of unit inspected}} = \frac{140}{400 \times 10} = 0.035$$

$$CL = n\bar{p} = 14$$

$$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1 - \bar{p})} = 25.025$$

$$LCL = n\bar{p} - 3\sqrt{n\bar{p}(1 - \bar{p})} = 2.975$$

Control Charts (CO5)



Process is out of control.

C-chart: is used when the no. of defect per Module are counted. Here C represent the no. of defects per Module (per item)

Construction of C-chart:

- Calculate $\bar{C} = \frac{\text{No.of defect in all samples}}{\text{total no.of samples}}$
- $CL = \bar{C}$

$$UCL = \bar{C} + 3\sqrt{\bar{C}}$$

$$LCL = \bar{C} - 3\sqrt{\bar{C}}$$

- If the no. of defects with in the UCL and LCL then process is in control. If it lie outside then process is out of control.

Example-1 Ten pieces of cloth out of different rolls of equal length contained the following defects: 1,3,5,0,6,0,9,4,4,3

Draw a control chart for the no. of defects and state whether the process is in state of statistical control.

$$\text{Sol: } \bar{C} = \frac{\text{No. of defect in all samples}}{\text{total no. of samples}} = \frac{35}{10} = 3.5$$

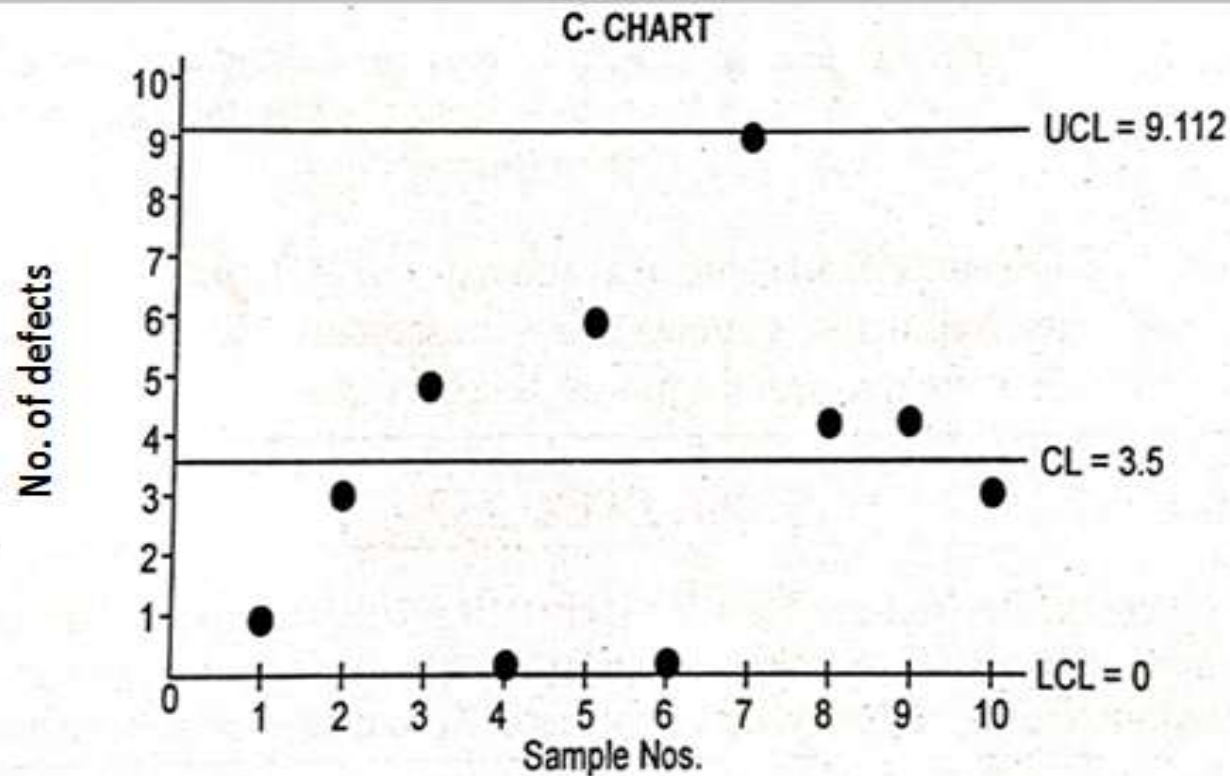
$$CL = \bar{C} = 3.5$$

$$UCL = \bar{C} + 3\sqrt{\bar{C}} = 9.112$$

$$LCL = \bar{C} - 3\sqrt{\bar{C}} = -2.112$$

here LCL is negative so LCL=0

Control Charts (CO5)



The above chart shows that all the plotted points are within the two control limits. This suggests that the process is in control.

Q1 The no. of defective needles of sewing machines has been given in the following table on the basis of daily inspection:

Day	1	2	3	4	5	6	7	8	9	10
Needle inspected	90	60	70	100	120	50	100	110	100	100
No. of defective needle	5	12	7	3	6	5	10	6	8	25

Prepare p-chart and state whether the production process is in control.

Ans: Process is not in control.

Q2: A process monitors the number of defects per day in the production Module and counts for 5 days are: [4, 6, 3, 5, 7]. Construct a C-chart.

Ans: The UCL is 11.71 and LCL is 0.

- **Self Made Video Link:**
- Test of hypothesis
- <https://youtu.be/h-NtACvrPfQ>
- Test of hypothesis for large samples-Z-test
- <https://youtu.be/a8MYgSRknml>
- Test of hypothesis for large samples-t test
- <https://youtu.be/msJXJgrY1t4>
- F-test
- <https://youtu.be/argjx9DJdxY>
- Chi square Test
- https://youtu.be/ayaa7_e1clA

Faculty Video Links, Youtube & NPTEL Video Links and Online Courses Details(CO5)

Self Made Video Link:

Statistical quality control & control chart for Variable

https://youtu.be/89bs12h_Cg0

Control chart for attributes

<https://youtu.be/q6BwAotTI04>

Faculty Video Links, Youtube & NPTEL Video Links and Online Courses Details(CO5)

Suggested video link

- Testing of hypothesis
- https://nptel.ac.in/content/storage2/courses/103106120/LectureNotes/Lec3_1.pdf

Weekly Assignment (CO5)

1. The following table gives the yields of four varieties of wheat grown in two plots:

Plots	Varieties			
	A	B	C	D
1	200	230	250	300
2	190	270	300	270
3	240	150	145	180

Is there any significant difference in the production of these varieties? (F value for (8,3)d.f = 4.07 and for (2,6)d.f = 5.14)

Ans. $F_{cal} = 0.385$; Significant

2. The following figures relate to producing in kg. of three varieties A, B and C of wheat shown in 12 plots :

Weekly Assignment (CO5)

A :- 14 16 18

B :- 14 13 15 22

C :- 18 16 19 19 20

Is there any significant difference in the production of these varieties?(F value for (2,9)d.f = 4.261)

Ans. $F_{cal} = 1.125$; Significant

3. The number of defects per Module in a Sample of 330 units of a manufactured product was found as follows

No. of defects	0	1	2	3	4
No. of units	214	92	20	3	1

Weekly Assignment (CO5)

(Given as 5% level of significance, the value of χ^2 for 1 d.f = 3.84).

Ans. $\chi^2_{cal} = 0.0295$; Significant

4. A milk producer's union wishes to test whether the preference pattern of consumers for its product is dependent on income levels. A random sample of 500 individuals gives the following data:

Income	Product			
	Preferred			
	Product A	Product B	Product C	Total
Low	170	30	80	280
Medium	50	25	60	135
High	20	10	55	85
Total	240	65	195	500

Weekly Assignment (CO5)

Can you conclude that the preference patterns are independent of income levels?(Given as 5% level of significance, the value of χ^2 for 4 d.f = 9.49).

Ans. $\chi^2_{cal} = 51.036$; Not Significant

5. Ten pieces of cloth out of different rolls of equal length contained the following number of defects: 1, 3, 5, 0, 6, 0, 9, 4, 4, 3. Draw a control chart for the number of defects and state whether the process is in a state of statistical control.

Ans. Process is in under statistical control

1. A sample of 20 items has mean 42 units and S.D. 5 units. It is a random sample from a normal population with mean 45 units. Then in test of hypothesis we get the following results? For this data $t_{\text{tab}} = 2.09$ at 5% level of significance.

- a) D.f. $\gamma = 19$
- b) $|t|_{\text{cal}} = 2.615$
- c) H_1 accepted
- d) There is significant difference between the sample mean and population mean.

2. A random sample of 900 members has mean 3.4cms. If it is reasonably regarded as a sample from a large population of mean 3.2cms and S.D. 2.3 cms. If the tabulated value for this

data is 1.96 at 5% level of significance. Then the following results are true.

- a) Use t-test for single mean.
- b) Use z-test for single mean.
- c) Use z-test for double mean.
- d) H_0 is accepted.

3. While testing the significance of difference of two sample means in case of small sample, how is degree of freedom calculated?

- a) $n_1 - 1$
- b) $n_1 + n_2 - 2$
- c) $(n_1 - 1) + (n_2 - 1)$
- d) $n_2 - 1$

4. What type of chart will be used to plot the number of defectives in the output of any process?

- a) $\bar{X}$ chart
- b) R chart
- c) c chart
- d) p chart

5. Consider a hypothesis H_0 where $\mu = 5$ against H_1 where $\mu > 5$. The test is?

- a) Right tailed
- b) Left tailed
- c) Center tailed
- d) Cross tailed

1. Pick out the correct option from Glossary-

I. z-test

II. F-test

III. t-test

IV. Chi Square test

A. Is used to test the significance for small sample

B. Is used to test the significance for large sample

C. Is also known as variance ratio test

D. Is used to test the independence of attributes

1. Pick out the correct option from Glossary-

- I. np-chart
 - II. p-chart
 - III. c-chart
 - IV. Control chart for variables
-
- A. Fraction defective chart
 - B. No. of defective chart
 - C. No. of defect chart
 - D. Mean chart

Old Question Papers (CO5)

[First Sessional Set-1 \(CSE,IT,CS,ECE,IOT\).docx](#)

[Second Sessional Set-2 \(CSE,IT,CS,ECE,IOT\).docx](#)

[Maths IV PUT.docx](#)

[Maths IV final paper_2022.pdf](#)

End Semester Question Paper

Printed Page:- 06

Subject Code:- BAS0303

Roll. No:

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA
(An Autonomous Institute Affiliated to AKTU, Lucknow)

B.Tech

SEM: III - THEORY EXAMINATION (2024- 2025)

Subject: Statistics & Probability

Time: 3 Hours

Max. Marks: 100

General Instructions:

IMP: Verify that you have received the question paper with the correct course, code, branch etc.

1. This Question paper comprises of **three Sections -A, B, & C**. It consists of Multiple Choice Questions (MCQ's) & Subjective type questions.
2. Maximum marks for each question are indicated on right -hand side of each question.
3. Illustrate your answers with neat sketches wherever necessary.
4. Assume suitable data if necessary.
5. Preferably, write the answers in sequential order.
6. No sheet should be left blank. Any written material after a blank sheet will not be evaluated/checked.

SECTION-A

20

1. Attempt all parts:-

1-a. Wheat crops badly damaged on account of rains is: (CO1, K2)

1

- (a) Cyclical movement
- (b) Random movement
- (c) Secular trend
- (d) Seasonal movement

End Semester Question Paper

- 1-b. Let the average of three numbers be 16. If two of the numbers are 8 and 12, then the remaining number is..... (CO1, K3) 1
- (a) 28
(b) 18
(c) 12
(d) 30
- 1-c. One card is drawn from a standard pack of 52 plying cards. Find the probability that it is either a king or a queen. (CO2, K3) 1
- (a) $\frac{1}{13}$
(b) $\frac{2}{13}$
(c) $\frac{3}{13}$
(d) None of these
-
- 1-d. If two events A and B are mutually exclusive, then the probability $P(A \cap B)$ is: (CO2, K2) 1
- (a) 0
(b) 1
(c) $P(A).P(B)$
(d) None of these

End Semester Question Paper

- | | | |
|------|---|---|
| 1-e. | In binomial distribution probability of success in each trial remains _____.(CO3. K1) | 1 |
| | <ul style="list-style-type: none"> (a) 0 (b) 1 (c) Constant (d) Not defined | |
| 1-f. | In Normal Distribution, Mean deviation about mean is (CO3, K1) | 1 |
| | <ul style="list-style-type: none"> (a) σ (b) $2\sigma/5$ (c) $4\sigma/5$ (d) $6\sigma/7$ | |
| 1-g. | The area of critical region depends on the size of.... (CO4, K1) | 1 |
| | <ul style="list-style-type: none"> (a) Type I error (b) Type II error (c) Test statistics (d) Sample | |
| 1-h. | In conducting one way analysis of variance --- test statistics would be used. (CO4,K1) | 1 |
| | <ul style="list-style-type: none"> (a) Z (b) T (c) Chi-Square (d) F | |

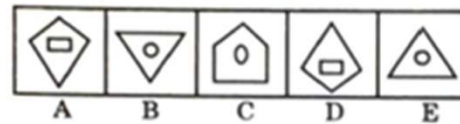
End Semester Question Paper

1-i. If $A = \{1,4,6\}$, $B = \{3,6\}$ and $C = \{3,4,6\}$ then $A \cap (B \cap C)$ is.... (CO5,K3) 1

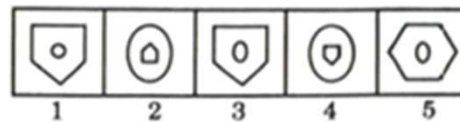
- (a) $\{3,4,5,6\}$
- (b) $\{4,6\}$
- (c) $\{1,4,6\}$
- (d) $\{6\}$

1-j. Select a figure from amongst the answer figure which will continue the same series as established by the five-figure problem. (CO5,K2) 1

Problem Figure



Answer Figure



- (a) 1
- (b) 2
- (c) 3
- (d) 5

End Semester Question Paper

2. Attempt all parts:-

- | | | |
|------|--|---|
| 2.a. | Prove that if two variables are independent, their correlation is zero but vice versa is not true. (CO1,K2) | 2 |
| 2.b. | Find the probability of getting 53 Sundays in a leap year. (CO2,K3) | 2 |
| 2.c. | A Binomial random variable X satisfies the relation $9P(X=4) = P(X=2)$, When $n=6$. Find value of $P(X=1)$. (CO3, K3) | 2 |
| 2.d. | Define an estimator in statistics. (CO4, K1) | 2 |
| 2.e. | Check whether the function $f: N \rightarrow N$ is defined by $f(x)=x^2+12$ is one-one or not. (CO5, K3) | 2 |

SECTION-B

30

3. Answer any five of the following:-

- | | | |
|------|--|---|
| 3-a. | The mean and standard deviation of the marks of 100 candidates was found to be 40 and 5.1, respectively. Later, it was discovered that a score of 40 was wrongly read as 50. Find out the correct mean and standard deviation respectively. (CO1,K3) | 6 |
| 3-b. | Calculate Spearman's rank correlation coefficient from the following data: (CO1, K3) | 6 |

End Semester Question Paper

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

- 3-c. A random variable X has the following distribution, Find the value of K. Also find Mean and variance (CO2, K3) 6

x	-2	-1	0	1	2	3
P(x)	0.1	K	0.2	2K	0.3	K

- 3-d. A bag X contains 2 white and 3 red balls and another bag Y contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag Y. (CO2, K3) 6

- 3.e. If 10% of the bolts produced by a machine are defective, determine the probability that out of 10 bolts chosen at random 6
- 1
 - None
 - at most 2 bolts will be defective. (CO3,K3)

- 3.f. A sample of 20 items has mean 42 units and S.D. 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units. (If the tabular value at 5% LOS for 19 d. f. is 2.09). (CO4, K3) 6

- 3.g. How many different words can be formed using all the letters of the word ALLAHABAD 6

- When the vowels occupy the even position.
- Both L do not occur together. (CO5,K3)

End Semester Question Paper

SECTION-C

50

4. Answer any one of the following:-

4-a. In a partially destroyed laboratory record of analysis of a correlation data, the 10

following results only are legible: (CO1,K3)

Variance of $x = 9$; Regression equations: $8x - 10y + 66 = 0$, $40x - 18y = 214$.

What were (a) the mean values of x and y (b) the standard deviation of y (c) the coefficient correlation between x and y .

4-b. Fit a second degree parabola by the method of least squares to the following data 10
. (CO1, K3)

X	0	1	2	3	4
Y	1	4	10	17	30

5. Answer any one of the following:-

5-a. State and prove Bayes theorem. 10

In a Neighbourhood, 90% children were falling sick due flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, find the child's probability of having flu. (CO2, K3)

End Semester Question Paper

5-b. Let the two dimensional continuous random variable (X,Y) has joint PDF given by 10

$$f(x,y) = \begin{cases} 6x^2y, & 0 < x < 1, 0 < y < 1 \\ 0 & , \text{ elsewhere} \end{cases}$$

Find (i) $P(0 < x < 3/4, 1/3 < y < 2)$ (ii) $P(x + y < 1)$. (CO2, K3)

6. Answer any one of the following:-

6-a. A manufacturer of envelopes knows that the weight of the envelopes is normally distributed with mean 1.9gm and variance 0.01 square gm. Find how many 10

envelopes weighing (i) 2gm or more

(ii) 2.1gm or more, can be expected in a given packet of 1000 envelopes?

Given that the area under the standard curve

between $z = 0$ and $z = 1$ is 0.3413,

between $z = 0$ and $z = 2$ is 0.4772. (CO3, K3)

6-b. 10

Four coins were tossed 200 times. The number of tosses showing 0,1,2,3 and 4 heads were found to be as under.

Fit a binomial distribution to these observed results. Find the expected frequencies. (CO3, K3)

No. of Heads	0	1	2	3	4
No. of Tosses	15	35	90	40	20

End Semester Question Paper

7. Answer any one of the following:-

7-a.

10

To test of significance of the variations of the retail prices in the commodity in three principal cities: Mumbai, Bangalore and Chennai. The four shops were chosen at random in each city and prices observed in INR were as follows:

Mumbai	16	8	12	14
Bangalore	14	10	10	6
Chennai	4	10	8	8

Do the data indicate that the prices in the three cities are significantly different? Given that the tabular value of F is 4.26 5% LOS with d.f. is (2,9). (CO4,K3)

7-b. From the following data, find whether hair color and gender are associated.

10

End Semester Question Paper

<i>Gender</i> ↓	<i>Colour</i>					
	Fair	Red	Medium	Dark	Black	Total
Boys	529	849	504	119	36	2100
Girls	544	677	451	97	14	1783
Total	1136	1526	955	216	50	3883

Given that the tabular value of χ^2 is 9.488 at 5%LOS with d.f. 4. (CO4, K3)

8. Answer any one of the following:-

8-a. Solve the following: (CO5,K3)

10

1. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 without repetition?
2. What is the sum of all five-digit numbers formed by 2, 3, 4, 5, 6 with repetition?

End Semester Question Paper

8-b.

Study the following table and answer the questions based on it
Expenditures of a Company (in Lakh Rupees) per Annum Over the given Years.

10

Year	Item of Expenditure				
	Salary	Fuel and Transport	Bonus	Interest on Loans	Taxes
1998	288	98	3	23.4	83
1999	342	112	2.52	32.5	108
2000	324	101	3.84	41.6	74
2001	336	133	3.68	36.4	88
2002	420	142	3.96	49.4	98

- Find the average amount of interest per year which the company had to pay during this period?
- The total amount of bonus paid by the company during the given period is approximately what percent of the total amount of salary paid during this period?
- Total expenditure on all these items in 1998 was approximately what percent of the total expenditure in 2002?
- The total expenditure of the company over these items during the year 2000 is?
- The ratio between the total expenditure on Taxes for all the years and the total expenditure on Fuel and Transport for all the years respectively is approximately? (CO5,K3)

Text Books

- N. P. Bali: A Textbook of Engineering Mathematics-IV, University Science Press.
- H. K. Dass: Introduction to Engineering Mathematics, S. Chand.
- S. Ross: A First Course in Probability, 6th Ed., Pearson Education India, 2002.
- W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968.

Reference Books

- P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).
- R.K. Jain and S.R.K. Iyenger: Advance Engineering Mathematics; Narosa Publishing House, New Delhi.
- J.N. Kapur: Mathematical Statistics; S. Chand & Sons Company Limited, New Delhi.
- D.N.Elhance, V. Elhance & B.M. Aggarwal: Fundamentals of Statistics; Kitab Mahal Distributers, New Delhi.

Thank You